INTRODUCTION TO ANIMATION

BCA-ELECTIVE

PROGRAM :1

Aim: To translate the image object using GIMP.

Algorithm:

Step 1: Start the program

Step 2: Initialize window/screen

Step 3: Set initial x-position of text

Step 4: Display text at current position

Step 5:Increase x-position gradually

Step 6: Clear screen and redraw text

Step 7: Draw a shape at initial position

Step 8: Apply translation (Δx, Δy)

Step 9: Redraw the shape at new position

Step 10: Delay for visibility

Step 11: Repeat translation until limit reached

Step 12: Stop the Program

1. Use the Move Tool
The primary method for translation is the **Move Tool**.

- **Activation:** Press the **M** key or select the **Move** icon from the **Toolbox**.
- **Selection:** In the **Tool Options** (usually below the Toolbox), choose what you want to move: **Layer**, **Selection**, or **Path**.
- **Execution:** Click and drag the object on the canvas to its new position. Hold **Shift** while dragging to constrain movement to a straight horizontal or vertical line.

2. Use Transform Tools for Advanced Translation

For translation combined with other changes (like scaling or rotating), use **Transform Tools** found under **Tools → Transform Tools**.

- **Unified Transform:** This versatile tool (Shortcut: **Shift + T**) allows you to move, scale, and rotate simultaneously by dragging different handles.
- **3D Transform:** Allows you to translate an object along the X, Y, and Z axes to create a perspective effect (Shortcut: **Shift + W**).
- **Handle Transform:** Lets you place "handles" on an object and drag them to shift its position or shape.

3. Translation via Selection
If the object is part of a larger layer, you must isolate it first:

1. Use a selection tool (e.g., **Fuzzy Select**, **Free Select**, or **Scissors Select**) to outline the object.
2. Go to **Edit → Cut** and then **Edit → Paste as New Layer** to turn it into an independent "object".
3. Use the **Move Tool** (M) as described above to translate the new layer.

Output:

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https://brush.ninja/create/animation-maker/

Program 2: Draw animated image using frames.

Aim : Draw an image using frame by frame using brush. Ninja tools.

Algorithm:

Step 1: Ninja Brush concept: fast, sharp, radial strokes with randomness (jitter)

Step 2: Polar coordinates (angle + radius)

Step 3: Random noise for dynamic brush effects

Step 4: Frame-based animation loop

Step 5: Visual effect: rotating, glowing, weapon-like brush strokes (ninja-style)

Step 6: Draw object at position (x_1, y_1)

Step 7: Draw object at (x_2, y_2)

Step 8: Repeat for N frames

Step 9: Define start frame

Step 10: nterpolate intermediate frames

Step 11: Frame by frame animation

Output:

Program 3: 3D software interface

Aim: Create an image using 3D software interface using GIMP.

Algorithms:

Step 1: Start GIMP

Step 2: Create a new image (600 × 400 px)

Step3: Select Rectangle Tool

Step 4: Draw a rectangular button

Step 5: Fill with base color

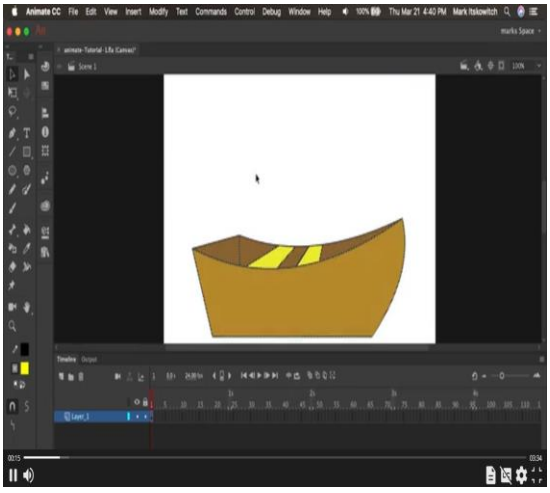
Step 6: Apply Gradient (light → dark)

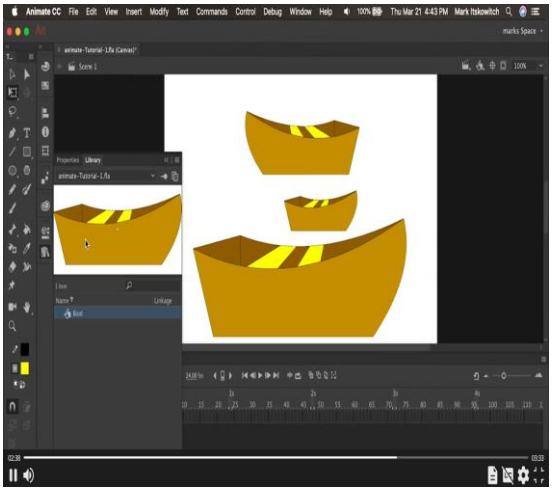
Step 7: Add Drop Shadow

Step 8: Add text on button

Step 9: Save the image

Output:





Program 4: Drawing tools, pen tools, create drawing in the frames.

Aim: To draw an image using the Pen Tool **and create animation using the** frame-by-frame algorithm **in** GIMP**.**

Algorithm:

Step 1: Start

Step 2: Open GIMP

Step 3: Create a new image File → New

Step 4: Select Paths Tool

Step 5: Click on canvas to place anchor point P1

Step 6: Click again to place P2

Step 7: Drag mouse to create curves

Step 8: Continue adding anchor points to form shape

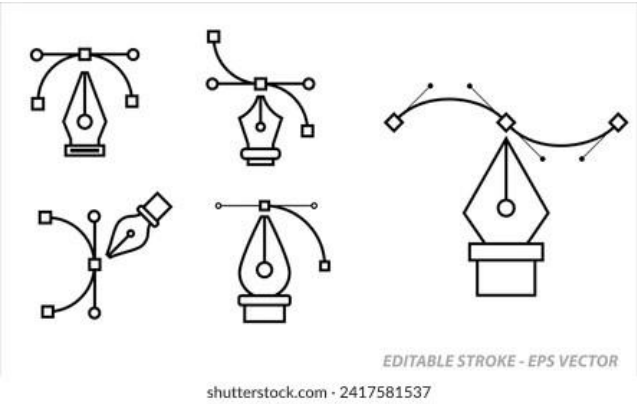
Step 9: Close the path

Step 10: Convert path to selection
Select → From Path

Step 11: Fill with color or stroke the path

Step 12: End

Output:



Program 5: Draw flipbook with animation principles

Aim: To create a flipbook-style animation using GIMP by applying basic animation principles such as timing, squash & stretch, anticipation, and ease in/ease out.

Algorithm:

Step 1: Create New File

Step 2: Create Frames Using Layers

Step 3: Draw Frame 1 (Anticipation)

Step 4: Draw Frame 2 (Falling – Fast Motion)

Step 5: Draw Frame 3 (Impact – Squash)

Step 6: Draw Frame 4 (Bounce Up – Stretch)

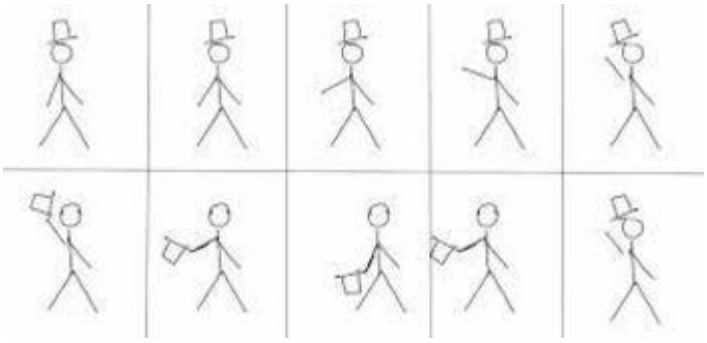
Step 7: Draw Frame 5 (Ease Out)

Step 8: Onion Skin Technique in GIMP

Step 9: Preview Animation in GIMP

Step 10: Export as Flipbook Animation

Output:



Program 6: Frame by frame animation of 10-20 sec

AIM: To create a 10–20 second frame-by-frame animation using GIMP

Algorithm:

Step 1: Plan the Animation

Step 2: Create New File in GIMP

Step 3: Draw First Frame

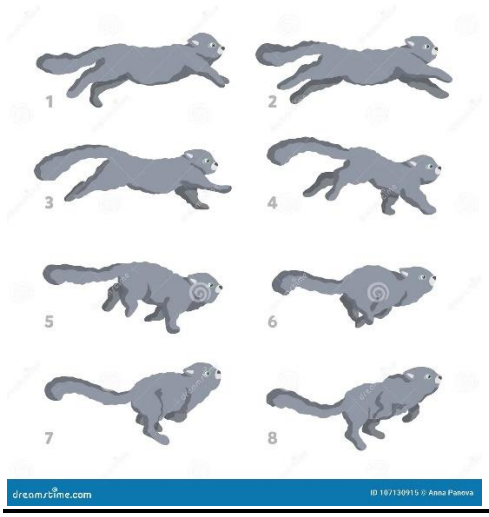
Step 4: Duplicate Frame (Layer)

Step 5: Create Frame-by-Frame Motion

Step 6: Add Smooth Motion (Animation Principles)

Step 7: Preview the animated image

Output:



Program 7: Create 20 second frame by frame color animation using GIMP

Aim: To create a 20-second frame-by-frame color animation **using** GIMP **by applying basic animation principles.**

Algorithm:

Step 1: Create a New Image

Step 2: Create Frames Using Layers

Step 3: Draw the First Frame

Step 4: Draw Remaining Frames

Step 5: Set Timing

Step 6: Preview Animation

Output:

Program 8: Creating simple animation with shape, classic & motion tweening.

Aim: To create animation with shape, classic & motion tweening.

Algorithm:

Used to change one shape into another shape (works only with raw shapes, not symbols).

Step 1: Start

Step 2: Open GIMP

Step 3: Create a new image (File → New)

Step 4: Create Layer 1 → Draw Shape A (e.g., Circle) using Ellipse Tool

Step 5: Duplicate Layer 1

Step 6: Modify the duplicated layer slightly (stretch/reshape)

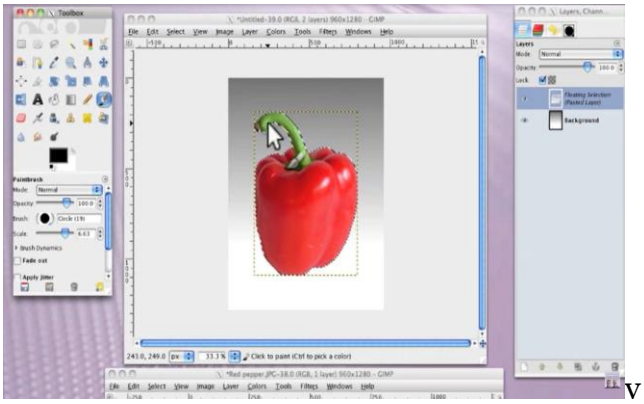
Step 7: Repeat duplication and gradual modification to reach Shape B (e.g., Rectangle)

Step 8: Arrange layers in correct sequence

Step 9: Open Filters → Animation → Playback to preview

Step 10: Export as GIF (File → Export As → .gif → As Animation)

Step 11: Stop



Program 9: A simple ball movement (rolling / bouncing) animation using GIMP.

Aim : To create a simple ball movement (rolling / bouncing) animation using GIMP.

Algorithm:

Step 1: Start

Step 2: Open GIMP

Step 3: Create a New Image

- **File → New**
- **Set width and height (e.g., 800 × 400 pixels)**

Step 4: Create Layer 1

- **Select Ellipse Tool**
- **Draw a circle (ball)**
- **Fill with color (e.g., red)**

Step 5: Duplicate Layer 1

- **Right-click layer → Duplicate Layer**

Step 6: Select Move Tool

- **Move the ball slightly to the right**

Step 7: Duplicate the layer again

Step 8: Repeat steps 6–7

- **Gradually move the ball frame by frame**

Step 9: (Optional – Bouncing Effect)

- **Move the ball downwards, then upwards in successive frames**

Step 10: Arrange all layers in correct sequence

Step 11: Preview animation

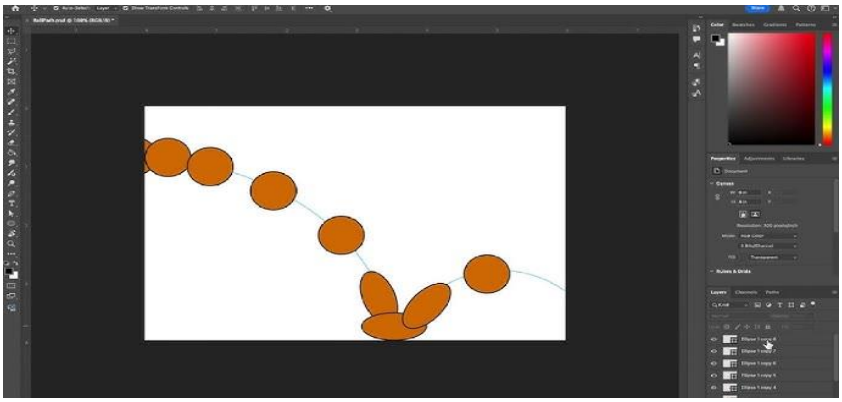
- **Filters → Animation → Playback**

Step 12: Export animation

- **File → Export As → ball_animation.gif**

- **Select As Animation**
- **Set frame delay (e.g., 100 ms)**

Step 13: End
Output:



Program 10: Character and create reusable symbols

Aim : To draw a character and create reusable symbols using various GIMP drawing tools.

Algorithm:

Step 1: Create a New File
Step 2: Sketch the Character Outline
Step 3: Draw Clean Outline
Step 1: Draw the Symbol
Step 2: Convert Symbol into Reusable Form
Step 4: Coloring the Character
Step 5: Add Details

Output:

Program 11: Human walk cycle

Aim: To create a basic human walk cycle using frame-by-frame animation in GIMP.

Algorithm:

Step 1: Start

Step 2: Open GIMP

Step 3: Create a New Image

- File → New (e.g., 800 × 600 pixels)

Step 4: Create Layer 1

- **Draw a stick-figure human (head, torso, arms, legs)**
- **This is the Contact Pose**

Step 5: Reduce layer opacity to 30% **(onion skin effect)**

Step 6: Create Layer 2

- **Draw the Down Pose**
- **Front leg bent, body lowered**

Step 7: Reduce opacity of Layer 2

Step 8: Create Layer 3

- **Draw the Passing Pose**
- **Back leg passes the front leg**

Step 9: Reduce opacity

Step 10: Create Layer 4

- **Draw the Up Pose**
- **Body lifted, stepping forward**

Step 11: Duplicate and mirror poses for the opposite leg

Step 12: Rename layers sequentially

- **Frame 1, Frame 2, … Frame 8**

Step 13: Preview animation

- **Filters → Animation → Playback**

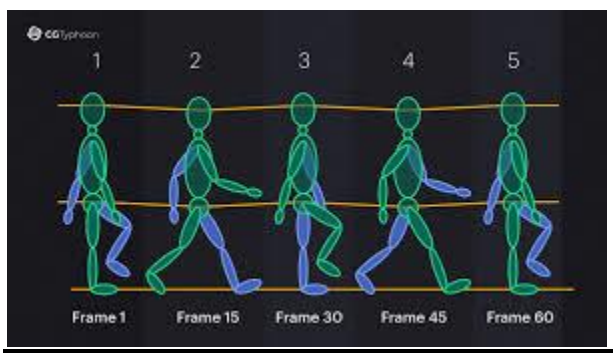
Step 14: Adjust timing (100–120 ms per frame)

Step 15: Export as animation

- **File → Export As → walk_cycle.gif**
- **Select** As **Animation**

Step 16: Stop

Output:



Program 12: Animal Walk cycle

Aim: To create a simple animal walk cycle **(e.g., dog / horse / cat) using** frame-by-frame animation **in GIMP.**

Algorithm:

Step 1: Start

Step 2: Open GIMP

Step 3: Create a **New Image**

- File → New (e.g., 900 × 500 pixels)

Step 3: Create Layer 1

- Draw a **simple animal side view** (head, body, four legs, tail)
- This is the **Contact Pose**

Step 4: Reduce Layer 1 opacity to 30% (onion skin)

Step 5: Create Layer 2

- Draw the **Down Pose**
- Body slightly lowered, weight on front leg

Step 6: Reduce opacity of Layer 2

Step 7: Create Layer 3

- Draw the **Passing Pose**
- Rear leg passes the standing leg

Step 8: Reduce opacity

Step 9: Create Layer 4

- Draw the **Up Pose**
- Body lifted, leg pushing off

Step 10: Duplicate frames and mirror leg positions for opposite side

Step 11: Rename layers sequentially

- Frame 1, Frame 2, … Frame 8

Step 12: Preview animation

- Filters → Animation → Playback

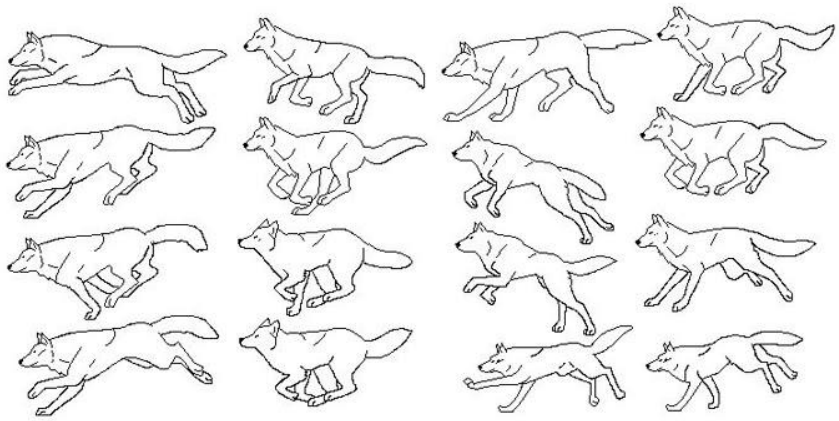
Step 13: Adjust frame delay (100–150 ms)

Step 14: Export animation

- File → Export As → animal_walk.gif
- Select **As Animation**

Step 15: Stop

Output:



Program13: Lip Synchronizing character mouth shape & dialogue

Aim: To synchronize a character's mouth movements with spoken dialogue using frame-by-frame animation in GIMP.

Algorithm:

Step 1: Start

Step 2: Open GIMP

Step 3: Create a **New Image**

- File → New (e.g., 800 × 600 pixels)

Step 4: Draw the character face on **Layer 1**

- Neutral mouth position (rest pose)

Step5: Prepare mouth shapes (phonemes) on separate layers:

- Closed mouth (M, B, P)
- Slight open (E, I)
- Wide open (A)
- Rounded mouth (O, U)
- Teeth visible (F, V)

Step 6: Import or listen to the dialogue audio

Step 7: Break the dialogue into phonemes/syllables

Step 8: Create a new layer for each frame

Step 9: Change the mouth shape according to the sound:

- **Assign correct mouth shape for each phoneme**

Step 10: Reduce opacity of previous frame (onion skin)

Step 11: Adjust mouth timing frame by frame

Step 12: Rename layers sequentially

- **Frame 1, Frame 2, ... Frame N**

Step 13: Preview animation

- **Filters → Animation → Playback**

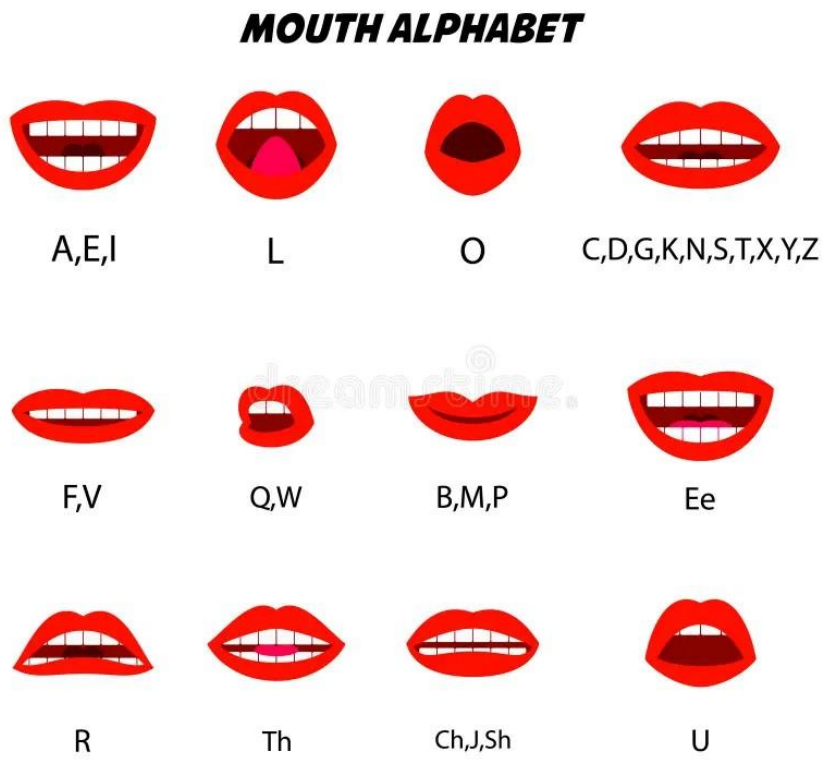
Step 14: Adjust frame delay (80–120 ms)

Step 15: Export animation

- **File → Export As → lip_sync.gif**
- **Select As Animation**

Step 16: Stop

Output:



Program 14 : Create short film by group (2) or individual